

Yiqing Zhi

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WORK EXPERIENCE

Saint-Gobain Abrasives

Lab/Quality Assistant Engineer Intern

Plattsville, ON

May 2026 – Present

- Led **quality testing** and **customer complaint investigations** on coated abrasives, analyzing grit performance, material condition, and process variation to drive corrective actions and quality decisions.
- Executed **Schiefer wear testing** (P80–P320 grits) by preparing conduits, running controlled RPM/load cycles, and quantifying wear through precise weight-change measurements.
- Applied **statistical analysis** (hypothesis testing, 95% CI) alongside **DA testing**, **fluid analysis**, and **CoA review** to validate grit classification and ensure traceable compliance with quality standards.

Linamar Corporation

Quality Engineer Intern

Guelph, ON

Sept 2025 – Dec 2025

- Performed **precision dimensional inspection** using CMM, gear analyzer, profile projector, and contour tracer to verify compliance with engineering drawings and tolerances.
- Analyzed measurement data using **SPC** and **Minitab**, identifying trends and reducing process variation through data-driven quality improvements.
- Supported **RCA/CAPA** and conducted **metallurgical testing** (Nital Etch, Rockwell, Brinell), aligning Control Plans and PFMEA with verified corrective actions.

PWO Canada Inc.

Process Engineering Assistant Intern

Kitchener, ON

Jan 2025 – April 2025

- Developed **Excel VBA tools** to streamline workflows, improving operator efficiency by **20%** while supporting stamping and tooling operations.
- Optimized manufacturing equipment using **SolidEdge** and 3D visualization tools, increasing press speed from **50 to 60 SPM** and boosting efficiency by **30%**.
- Reduced operational waste through **time studies**, **SAP-based inventory optimization**, and **Lean/Six Sigma** initiatives, lowering discrepancies by **10%** and cutting monthly PPE costs.

University of Waterloo Formula Electric

Chassis Responsible Engineer

Waterloo, ON

Sept 2024 – Present

- Designed a **compact electronics enclosure and dashboard** in SolidWorks, integrating waterproofing, mounting constraints, and system requirements for race conditions.
- Prototyped and validated designs using **3D printing** (PETG) and supported chassis fabrication via **machining**, **waterjet cutting**, and welding fixture setup.
- Produced detailed **CAD models and DXF drawings** under tight mechanical constraints, collaborating cross-functionally to ensure manufacturability and performance.

PROJECTS

Circadian Rhythm Regulating Device

- Designed and integrated a **mechanical wake-up system** using a rotating arm with tactile feedback and a gravity-fed water mechanism, controlled via an **EV3 microcontroller** to eliminate snooze behavior.
- Engineered a **gear train and bearing-supported shaft system**, optimizing alignment and minimizing friction/backlash to achieve smooth, low-torque rotational actuation.
- Developed **embedded control logic** in RobotC to synchronize motor motion, valve actuation, and sensor feedback, enabling reliable **multi-phase system operation**.

EDUCATION

University of Waterloo

Bachelor of Applied Science in Mechanical Engineering - Option in Computing

Waterloo, ON

Sept 2024 – Present (2029)

SKILLS

Quality & Testing: Schiefer, DA, Fluid Analysis, CoA, CMM, Nital Etch, Rockwell/Brinell, SPC (Minitab).

Quality Systems: Control Plan, PFMEA, RCA/CAPA, Process Capability, Traceability.

Design & Manufacturing: SolidWorks, SolidEdge, AutoCAD, Tech Drawing, 3D Printing, Lathe, Drill Press.

Software & Data: VBA, SAP, Python, MATLAB, C++, RobotC.